

Practical session 1: Regression for data analysis

The focus of this exercise is on **Feature Selection using Logistic Regression**. Feature selection is an important step in building efficient and interpretable models, as it helps identify the most relevant features for prediction and reduces the complexity of the model.

The aim is to analyze the features on the colon dataset.

Steps for Feature Selection using Logistic Regression

1. Train a Logistic Regression Model:

- Apply **Logistic Regression** to the dataset, using all available features (genes) to predict the target variable ("normal" or "cancer").
- Use the normalized training and test datasets.

2. Evaluate Feature Importance:

- After training, access the coefficients (`coef_`) of the Logistic Regression model.

3. Sort Features by Their Impact:

- Create a table (e.g., using pandas DataFrame) to display the coefficients of each feature.
- Sort the features by the absolute value of their coefficients to identify the most important features.

4. Visualize Feature Importance:

- Create a bar plot of the coefficients sorted by magnitude.
- This visualization will help you understand which features have the highest impact on the classification.

5. Select the Most Important Features:

- Based on the sorted coefficients, select the top N features (e.g., top 3 features with the highest absolute coefficient values).

6. Retrain the Model Using Only the Selected Features:

- Retrain the Logistic Regression model, but this time only using the top N selected features (those with the highest absolute coefficients).
- This step will help assess whether using fewer features can maintain or improve model performance, leading to a more interpretable and efficient model.

7. Model Performance with Selected Features:

- Train and evaluate Logistic Regression using only the selected features.
- Compare the performance of the retrained model with the original model (which used all features) to see if there is any change in performance.

8. Correlation Analysis of Selected Features:

- To check if the top selected features are highly correlated with each other, plot a correlation matrix or a pairplot.
- If features are strongly correlated, they may provide redundant information, and you could consider removing one of them for better model performance.

Now, let's extend the feature selection methodology to a different dataset in the business or finance sector. Choose a dataset related to banking, finance, or customer behavior, such as predicting customer churn, loan approval, or fraud detection. You will follow the same steps as described previously, but this time on your chosen dataset. Interpret the different results obtained.